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CurricuLLM: Full test dataset results

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In this supplementary, we provide the full annotations of the hold-out test set, consisting of 3 courses for each of the curricula analyzed in (KTH, NTU, CMU). The results for the Building Networked Systems Security (KTH) course are provided earlier in the paper in Table 4. The rest of the course annotations and results can be found in Table 2: Ethical Hacking (KTH), Language-Based Security (KTH), Network Security (NTU), Software Security (NTU), Privacy Preserving Technologies & Security in AI (NTU), Advanced Real-World Networks (CMU), Networked Systems Course (CMU), Information Security Policy and Management (CMU).

For each course analyzed, we derived topics from the course descriptions using PreprocessLM, labeled them using ClassifyLM and present the final result of this process under the column CurricuLLM in the table. Then, Control Group A, consisting of 34 cybersecurity experts, and a group of three cybersecurity experts previously familiar with the CSEC2017 framework (noted as ‘Expert Group X’) were consulted to independently label the topics. The numbers listed in the tables down below represent these annotations, where the corresponding KAs for each number can be found in Table 1.

Color	Number	Knowledge Area (KA)	Concise Description
	0	Miscellaneous	Computer Science, Business & Law, Communication & Networking, Information Technology, Cyberspace Practice, Pedagogy, Intelligence
	1	Data Security	Basic cryptography concepts, Digital forensics, End-to-end secure communications, Data integrity & authentication, Information storage security
	2	Software Security	Fundamental design principles, Security requirements, Implementation issues, Static & dynamic testing, Configuring & patching, Ethics
	3	Component Security	Vulnerabilities of system components, Component lifecycle, Secure component design principles, Supply chain management security, Security testing, Reverse engineering
	4	Connection Security	Systems, architecture, models, standards, Physical component interfaces, Software component interfaces, Connection attacks, Transmission attacks
	5	System Security	Holistic approach, Security policy, Authentication, Access control, Monitoring, Recovery, Testing, Documentation
	6	Human Security	Identity management, Social engineering, Awareness & understanding, Social behavioral privacy & security, Personal data privacy & security
	7	Organizational Security	Risk management, Governance & policy, Laws, ethics, & compliance, Strategy & planning
	8	Societal Security	Cybercrime, Cyber law, Cyber ethics, Cyber policy, Privacy

TABLE 1: Cybersecurity Concept Classification. This table lists the eight CSEC2017 KAs [1] (classes 1–8) and a Miscellaneous class (0) from [2]. KA numbers and colors are used consistently throughout the paper for readability.

TABLE 2: Topics extracted from all courses in the hold-out test set, annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Ethical Hacking Course (KTH)							
Ethical Hacking	1,6	6,8	2,5,6	0–8	2,7,8	0,2–6	8
Network and vulnerability scanning	4	3,4,5,7	1,4	4,5	5	0,4	4
Exploit development platforms	2,7	3	2,3	2	2	2,3,4	2
Command and control	0	2	2,3,6	0	7	0,2,4,5	0
Password cracking	1	5,6	1,2,3	0,1	5	1,2,6	6
Independent attack project	1,3	–	7	0	5	0	5
Virtual environment setup	4	7	1,2	4,5	0	0	4
Language-Based Security Course (KTH)							
Language-Based Security	1,6	2	–	0,2	6	0,1,2,6,7	5
Fundamental principles, models and concepts for computer security	0	2	1,4,5,7,8	0–5	1,3	0–8	5
Software security by information flow control	3	4	1,2,4	2	2	2	2
Web application and database security	2	2	1,4,5,6,7	0,2	2,5	1,2,4,5,7	7
Security for mobile applications	5,6	2	1,3,4,5,6	4	2,4	2,5,6	7
Hot topics in computer security	1	–	0,5,8	0–8	1,3,4,5	0–8	7
State-of-the-art in programming language for security	1,6	2	2,3,4	2	2	2	7
Network Security Course (NTU)							
Network Security	5	3,4,5,7	1,2,5,6,7	4	4,5	4	7
Introduction to Networking	5	1,2,3,4,5	4	0	0	4	4
Security in Networks	5	3,4,5,7	1,2,5,6	4	4,5	4	7
Network Programming	5	4,5	2	0,4	2,4	2,4	7
Packet Trace Analysis with Wireshark	5	4	1,4,7	4	4	3,4	4
TLS and Security in Networked Systems	5	1,2,4,5	1,4,5	4	4,5	4	4
Software Security Course (NTU)							
Software Security	3,4,5,7	–	2	2	2	2	2
Challenges in software security	1–5	8	2	2	2	2	2
Principles of secure software development	3–5,7	2	0,2,5	2	2	2	2
Mechanisms for secure coding	4	2,5	5	2	2	2	2
Tools for software security	2,4	1	2	2	2	2	2
Causes of vulnerabilities	1,2,4,5	3,6	0,6,7	0–6	2–5	0–8	7
Strategies to avoid vulnerabilities	1–3	2	3,5,7	0–6	2,3	2,5,6,7,8	7
Defenses against attacks	1–5,7	1,6	3,5,7	0–6	5,7	0–8	5
Secure programming practices	2	3	2,3,5,6,8	2	2	2	2
Language-specific secure coding	2,4	3	2,3,5,7	2	2	2	2
System-level defenses	1,2,4,5	5	1–5	5	5	5	5
Architectural approaches for security	1,3–5,7,8	2,4	0,3–8	4	5	0,5,6,7,8	2
Run-time enforcement techniques	2–6	1,5	0,4,6–8	4	2,3	2,3,5	7
Software analysis tools	1,2,4,6,7	2	0–2,4–8	2	0,2	0,2,3	2
Vulnerability detection methods	1–3,5–8	3	0,2,5–8	3	5	0,2–6	2,3

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TABLE 2 – continued from previous page

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Application of security tools in scenarios	0,2,5–8	–	0–2,5–8	3	5	0	7
Privacy Preserving Technologies & Security in AI Course (NTU)							
Privacy Preserving Technologies & Security in AI	1,6	0–3,5,7,8	0–2,6–8	0,2,3,6	5,6,8	0,1,6	1,6–8
Fully Homomorphic Encryption	1	0,1,4	1,3,5	1	1	1	1
Secure Multi-computation	1,2,4	0,2,3,5,7	1,3,5,7	1	1	1	1
Searchable Encryption	1	0,1,4	1,2,5	1	1	1	1
Data Sharing in Machine Learning	1	0,4,7,8	0–2,4,5,7	0,1	0	1,6	4
Advanced Real-World Networks Course (CMU)							
Advanced Real-World Networks	4,5	1–8	1,4,5	0,4	0	4	4
4G and 5G network infrastructures	4	1–8	0,3–5	0,4	0	1,4,5	4
IPv6	2,5	0	0,1,4	4	0	4	4
SDN and VFN	2	1,2,4,7	0,1,2,3,5	4	0	2,4	4
Data centers	1	1,3–6	0–7	0	0	1–5	1
Mesh networks	3,4	1–5	0,4	4	0	4	4
Embedded networks	3,4	1–8	0,4	4	0	2,4	4
Security in Networked Systems Course (CMU)							
Security in Networked Systems	1	1–5	1	4	4,5	4	7
Network and transport-layer attacks and defenses	4	1–5	4	4	4,5	4	4
Network intrusion detection	5,6,7	1–5	1,4	4	4,5	4	4
Denial of service (DoS) and distributed denial-of-service (DDoS) detection and reaction	4,5,8	4	1,4	4	4,5	4	4
Worm and virus propagation	2,3	6	2,3,5,7	4	2,4,5	0,2–6	1
Tracing the source of attacks	0–8	0–2,4–6–8	1,2	0–6	5	0,2,4,5	1
Traffic analysis	4	0,1,4,8	4	4	1,4	0,4	7
Techniques for hiding the source or destination of network traffic	1,2,5	0,1,4,6,8	2,4,5	4	4	4	4
Secure routing protocols	4	0,1,2,4,8	4	4	4	4	4
Content poisoning attacks	1,2,3	1–4	1	4	2–4	2,4	1
Advanced techniques for reacting to network attacks	4,8	0–6	0,6–8	4	4,5	0,4	4
Information Security Policy and Management Course (CMU)							
Information Security Policy and Management	0,1,5,6,8	7,8	7	7	5,7	0,7,8	7
Security marketplace overview	0,2,8	3	0	7	5,7	0	7
Decision making with multiple parties involved	5,7,8	6,8	8	7	0	0,1,7,8	7,8
Role of policy in information security	1,7	8	7	7,8	5,7,8	7	7,8
Intra-organization policies	4,7	4,8	3,5,6	7	0	7	7
Market and competition impact on security provision	0,7,8	1,2	8	7	5,7,8	0,7,8	7
Causes of market failure (externalities)	0,5,7,8	0,7	6–8	0,7	0	0,7,8	7
Policy tools to mitigate market failure	0,5	0,8	3,7,8	7	0	0,7	7,8

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TABLE 2 – continued from previous page

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Key laws and regulations on product liability and security standards	5	3,7,8	1,2	7,8	8	0,3,4,5,7,8	7
Overview of security industry trends, technologies, vendor and user strategies	0	0–5	0,4	7	8	0	7
Managerial and policy issues in information security provision	5,7	0,1,3,6	0,2	7	7,8	7	7
When and how policy intervention is needed	–	0,7,8	6,7	7,8	8	0,7,8	7,8

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